Practical No. 09

Q.1] Write a C program for factorial calculation using for loop and test it.

```
#include <stdio.h>
#include <conio.h>
int main()
{
   int n, i, f = 1;
   printf("Enter number: ");
   scanf("%d", &n);
   if (n < 0)
   {
      printf("The factorial is not defined");
   }
   else{
      for (i = 1; i <= n; i++)
      { f = f * i;
      }
      printf("Factorial of %d is %d", n, f);
   }
   return 0;
}
```

Test Cases :-

Test ID	Objective	Input (Steps)	Expected Output	Actual Output	Status
TC_01	Verify factorial calculation for a positive number	Enter n = 5	Factorial of 5 is 120	Factorial of 5 is **120**	Pass
TC_02	Verify program behavior for valid number without error	Enter n = 6	Factorial of 6 is 720	Factorial of 6 is **720**	Pass
TC_03	Verify factorial calculation for a small positive number	Enter n = 3	Factorial of 3 is 6	Factorial of 3 is **6**	Pass
TC_04	Verify factorial of zero	Enter n = 0	Factorial of 0 is 1	Factorial of 0 is **1**	Pass
TC_05	Verify handling of negative input	Enter n = -3	Error message: *Factorial is not defined*	Error message: *Factorial is not defined*	Pass

Q.2 Design five new test cases for do…while to test condition variations.

Test Case ID	Objective	Input (Steps)	Expected Output	Actual Output	Status
TC_01	Verify loop executes minimum once for valid input	Enter i = 5	Output displays 5	Output displays 5	Pass
TC_02	Validate loop exit condition for out-of-range input	Enter i = 6	Error message: *Invalid input*	Output displays 6	Fail
TC_03	Verify loop starts from correct initial value	Enter i = 2	Output: 2 3 4 5	Output: 2 3 4 5	Pass
TC_04	Verify handling of negative input	Enter i = -1	Error message: *Invalid input*	Error message displayed	Pass
TC_05	Detect incorrect boundary condition	Enter i = 7	Error message: *Invalid input*	Error message displayed	Pass

Q.3 Create a decision table for a switch case structure.

Test Case ID	Objective	Input (Steps)	Expected Output	Actual Output	Status
TC_01	Verify loop executes for lowest valid boundary value	Enter i = 1	Output: 1 2 3 4 5	Output: 1 2 3 4 5	Pass
TC_02	Verify loop stops at upper limit	Enter i = 5	Output ends at 5	Output ends at 5	Pass
TC_03	Verify program behavior for null/empty input	Press Enter without input	Error message: *Invalid input*	Error message displayed	Pass
TC_04	Verify loop does not execute for invalid high value	Enter i = 10	Error message: *Invalid input*	Error message displayed	Pass
TC_05	Verify program handles non-numeric input	Enter i = a	Error message: *Invalid input*	Error message displayed	Pass